

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
Department of Geosciences and CGISS, Boise State University
Environmental Research Building (ERB) 1104 (lab), 3155 (office)
1910 University Drive
Boise, ID 83725

(208) 426-1416 (office)
(303) 859-3106 (cell)
hpmarshall@boisestate.edu
cgiss.boisestate.edu/~hpm

Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington
Thesis: *Wet snow densification*
- 1994 Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Professional Experience

- 08/2013-present Associate Professor, Dept. of Geosciences, Director of CryoGARS, Boise State University
- 06/2021-present Leadership team member, NASA SnowEx Mission
- 10/2021-present Avalanche Instructor and Senior Advisor, Avalanche Science LLC
- 06/2018-08/2021 Project Scientist, NASA SnowEx Mission (20+ organizations, 100+ participants)
- 08/2016-08/2022 Senior Project Staff Engineer, ATA Aerospace, LLC
- 07/2007-03/2020 Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers
Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire
- 08/2008-07/2013 Assistant Professor, Department of Geosciences and Center for Geophysical
Investigation of the Shallow Subsurface (CGISS), Boise State University
- 01/2005-08/2011 Chair of Research Committee, American Avalanche Association
- 04/2006-08/2008 Research Associate, Inst. of Arctic and Alpine Research (INSTAAR)
University of Colorado at Boulder
- 12/2004-04/2006 Research Geophysical Engineer (GS-11), Student Career Experiences Program (SCEP)
U.S. Army Corps of Engineers CRREL, Hanover, New Hampshire & Fairbanks, Alaska
- 01/2004-07/2004 Invited visiting scientist, Swiss Federal Institute for Snow and
Avalanche Research (SLF), Davos, Switzerland
- 07/2000-12/2005 Research Assistant, INSTAAR, University of Colorado at Boulder
- 11/2002-12/2003 Avalanche Instructor, Alpine World Ascents, Boulder, Colorado
- 06/1995-07/2000 Undergraduate Research Assistant, Glaciology Group, University of Washington
- 08/1999-03/2000 Volunteer high school teacher, Old Crow, Yukon Terr., Canada
- Summers 1993-94 Roofer, Seattle, WA

Research Interests

Snow science and glaciology, using geophysics, engineering, and remote sensing
Spatial variability of snow, snow and ice mechanics, snow slope stability, meltwater dynamics

Computational statistics, geostatistics, compressed sensing, instrumentation development

Honors and Awards

2020	BSU Faculty Excellence Award
2019	NASA Instrument Incubator Program Award, NASA, 1st in Idaho history
2019	2018 WRR Editors' Choice Award: Hedrick et al., 2018
2018	NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
2017	Incoming Chair of the International Snow Working Group for Remote sensing (ISWGR)
2012	Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
2010-13	New Investigator Program Award, NASA, 1st in Idaho history
2010	Young Scientist Award, Cryosphere Section, American Geophysical Union
2009	Top Reviewer Award, Cold Regions Science and Technology (CRST)
2008	U.S. Young Scientist Award, International Union of Radio Science (URSI)
2002-05	N.A.S.A. Earth Systems Science Graduate Fellowship
2003	Backcountry Access Avalanche Education Support Grant
2003	Arctic Research Consortium of the U.S. (ARCUS) Award for Arctic Research Excellence, Hon. Mention
2002	University of Colorado Graduate School Student Travel Grant
2001-02	N.S.F. K-12 Engineering Outreach Fellowship
2002	Beverly Sears Graduate Research Grant (University of Colorado)
2001	American Avalanche Association Graduate Research Grant
2000	Li Tze Fong Memorial Fellowship, University of British Columbia (declined)
1999	University of Washington Bonderman Honors Travel Fellowship
1999	Volunteer of the Year, Arctic Canada, Frontiers Foundation
1997-99	University of Washington Mary Gates Undergraduate Research Grant
1999	Featured speaker, University of Washington Undergraduate Research Symposium
1995-98	N.A.S.A. Space Grant Undergraduate Research Program
1999,2001	N.S.F. Graduate Fellowship Program - Honorable Mention
1994-95,97-99	Dean's List, University of Washington
1994-95	U.W. Undergraduate Scholar Award (1994-95)
1994	Washington Athletic Club Student Athlete of the Year, Nathan Hale High School

Refereed Publications [peer-reviewed publications=70, h-index=28, i10-index=53, citations=2413]

- 2022 Lievens, H., Brangers, I., **H.-P. Marshall**, Jonas, T., Olefs, M., and DeLannoy, G. (2022). Sentinel-1 snow depth retrieval at sub-kilometer resolution over the European Alps. *The Cryosphere*, 16(1):159–177, doi.org/10.5194/tc-16-159-2022
- McGrath, D., Bonnell, R., Zeller, L., Olsen-Mikitowicz, A., Bump, E., Webb, R., and **H.P. Marshall** (2022). A time-series of snow density and snow water equivalent observations derived from the integration of gpr and uav sfm observations. *Frontiers in Remote Sensing*, 3:886747, doi:10.3389/frsen.2022.886747
- Liu, J., Enderlin, E. M., **H.-P. Marshall**, and Khalil, A. (2022). Synchronous retreat of south-east greenland's peripheral glaciers. *Geophysical Research Letters*, 49(13):e2022GL097756, https://doi.org/10.1029/2022GL097756
- Lund, J., Forster, R., Deeb, E., Liston, G., Skiles, M., and **H.P. Marshall** (2022). Interpreting sentinel-1 sar backscatter signals of snowpack surface melt/freeze, warming, and ripening, through field measurements and physically-based snowmodel. *Remote Sensing*, 14:4002, https://doi.org/10.3390/rs14164002
- Tsang, L., Durand, M., Derksen, C., Barros, A. P., Kang, D.-H., Lievens, H., **H.-P. Marshall**, Zhu, J., Johnson, J., King, J., Lemmetyinen, J., Sandells, M., Rutter, N., Siqueira, P., Nolin, A., Osmanoglu, B., Vuyovich, C., Kim, E. J., Taylor, D., Merkouriadi, I., Brucker, L., Navari, M., Dumont, M., Kelly, R., Kim, R. S., Liao, T.-H., and Xu, X. (2022 (accepted)). Review article: Global monitoring of snow water equivalent using high frequency radar remote sensing. *The Cryosphere*, pages https://doi.org/10.5194/tc-2021-295
- Hojatimalekshah, A., Gongora, J., Enterkine, J., Glenn, N. F., Caughlin, T., and **H.P. Marshall** (2022 (accepted)). Lidar and machine learning reveal forest structural controls on snowpack. *Frontiers in Ecology and the Environment*
- Lemmetyinen, J., Cohen, J., Kontu, A., Hannula, J. V. H., Merkouriadi, I., Scheiblaue, S., Rott, H., Nagler, T., Ripper, E., Elder, K., **H.P. Marshall**, Fromm, R., Adams, M., Derksen, C., King, J., Meta, A., Coccia, A., Rutter, N., Sandells, M., Macelloni, G., Santi, E., Leduc-Leballeur, M., Essery, R., Menard, C., and Kern, M. (2022 (accepted)). Airborne snowsar data at x- and ku- bands over boreal forest, alpine and tundra snow cover. *Earth Syst. Sci. Data*, pages https://doi.org/10.5194/essd-2021-239
- Brangers, I., DeLannoy, G., Lievens, H., and **H.P. Marshall** (2022). Tower based C-band radar observations of the snowpack. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*
- Adebisi, N. and **H.P. Marshall** (2022). Snow depth retrieval from L-band data based on repeat pass InSAR techniques. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*
- Enderlin, E., Elkin, C., Gendreau, M., **H.P. Marshall**, O'Neel, S., McNeil, C., Sass, L., and Florentine, C. (in review). Uncertainty of icesat-2 atl06- and atl08-derived snow depths for glacierized and vegetated mountain regions. *Remote Sens. Environ.*

2021

Meehan, T., Hojatimalkeshah, A., **H.P. Marshall**, Deeb, E., O’Neel, S., McGrath, D., Webb, R., Bonnell, R., Raleigh, M., Hiemstra, C., and Elder, K. (in review). Spatially distributed snow water equivalent from ground-based and airborne sensor integration at grand mesa, colorado, usa. *Water Resour. Res.*

H.-P. Marshall, Deeb, E., Forster, R., Vuyovich, C., Elder, K., Hiemstra, C., and Lund, J. (2021). L-band InSAR depth retrieval during the NASA SnowEx 2020 Campaign: Grand Mesa, Colorado. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 625–627, DOI:10.1109/IGARSS47720.2021.9553852

Lewis, G., Osterberg, E., Hawley, B., **H.-P. Marshall**, Meehan, T., Graeter, K. A., McCarthy, F., Overly, T., Thundercloud, Z., Ferris, D., Koffman, B., and Dibb, J. (2021). Atmospheric blocking drives recent albedo change across the western Greenland ice sheet percolation zone. *Geophysical Research Letters*, 48(10):e2021GL092814

Johnson, J., Anderson, J., **H.-P. Marshall**, Havens, S., and Watson, L. (2021). Snow avalanche detection and source constraints made using multiple infrasound sensor arrays. *Journal of Geophysical Research-Earth Surface*, 126(3):e2020JF005741

Liu, J., Enderlin, E. M., **H.P. Marshall**, and Khalil, A. (2021). Automated detection of marine glacier calving fronts using the 2d wavelet transform modulus maxima (wtmm) segmentation method. *IEEE Transactions On Geoscience and Remote Sensing*, 59(11):9047–9056, doi:10.1109/TGRS.2021.3053235

Meehan, T. G., **H.P. Marshall**, Bradford, J. H., Hawley, R. L., Overly, T. B., Lewis, G., Graeter, K., Osterburg, E., and McCarthy, F. (2021). Historical surface mass balance reconstruction 1984-2017 from greentracs multi-offset ground-penetrating radar. *Journal of Glaciology*, 67(262):219–228, doi:10.1017/jog.2020.91

Hojatimalekshah, A., Uhlmann, Z., Glenn, N. F., Hiemstra, C. A., Tennant, C. J., Graham, J. D., Spaete, L., Gelvin, A., **H.P. Marshall**, McNamara, J., and Enterkine, J. (2021). Tree canopy and snow depth relationships at fine scales with terrestrial laser scanning. *The Cryosphere*, 15(5):2187–2209, doi:10.5194/tc-15-2187-2021

Kim, R. S., Kumar, S., Vuyovich, C., Houser, P., Lundquist, J., Mudryk, L., Durand, M., Barros, A., Kim, E. J., Forman, B. A., Gutmann, E. D., Wrzesien, M. L., Garnaoud, C., Sandells, M., **H.P. Marshall**, Cristea, N., Pflug, J. M., Johnston, J., Cao, Y., Mocko, D., and Wang, S. (2021). Snow Ensemble Uncertainty Project (SEUP): Quantification of snow water equivalent uncertainty across north america via ensemble land surface modeling. *The Cryosphere*, 15:771–791, doi:10.5194/tc-15-771-2021

Webb, R. W., Marziliano, A., McGrath, D., Bonnell, R., Meehan, T. G., Vuyovich, C., and **H.-P. Marshall** (2021). In situ determination of dry and wet snow permittivity: Improving equations for low frequency radar applications. *Remote Sensing*, 13:4617, DOI:10.3390/rs13224617

Bonnell, R., McGrath, D., Williams, K., Webb, R., Fassnacht, S. R., and **H.-P. Marshall** (2021). Spatiotemporal variations in liquid water content in a seasonal snowpack: Implications for radar remote sensing. *Remote Sensing*, 13:4223, DOI:10.3390/rs13214223

- Lievens, Hans and Brangers, Isis and Marshall, Hans-Peter and Jonas, Tobias and Olefs, Marc and De Lannoy, Gabri lle (2021). Observing snow depth at sub-kilometer resolution over the european alps from sentinel-1. *2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS*, pages 618–621, doi:10.1109/IGARSS47720.2021.9553420
- Tsang, L., Durand, M., Derksen, C., Barros, A. P., Kang, D.-H., Lievens, H., **Marshall, H.P.**, Zhu, J., Johnson, J., King, J., Lemmetyinen, J., Sandells, M., Rutter, N., Siqueira, P., Nolin, A., Osmanoglu, B., Vuyovich, C., Kim, E. J., Taylor, D., Merkouriadi, I., Brucker, L., Navari, M., Dumont, M., Kelly, R., Kim, R. S., Liao, T.-H., and Xu, X. (2021). Review article: Global monitoring of snow water equivalent using high frequency radar remote sensing. *The Cryosphere Discussions*, pages doi.org/10.5194/tc-2021-295
- 2020 Hedrick, A. R., Marks, D., **H.P. Marshall**, McNamara, J., Havens, S., Trujillo, E., Sandusky, M., Robertson, M., Johnson, M., Bormann, K. J., and Painter, T. H. (2020). From drought to flood: A water balance analysis of the tuolumne river basin during extreme conditions (2015 – 2017). *Hydrol. Process.*, 34(11):2560–2574
- Webb, R., Raleigh, M., McGrath, D., Molotch, N., Elder, K., Hiemstra, C., Brucker, L., and **H.P. Marshall** (2020). Within-stand boundary effects of snow water equivalent distribution in forested areas. *Water Resources Research*, 56:e2019WR024905
- Lund, J., Forster, R. R., Rupper, S. B., **H.P. Marshall**, Deeb, E. J., and Hashmi, M. (2020). Mapping snowmelt progression in the upper indus basin with synthetic aperture radar. *Frontiers in Earth Science*, 7:318, doi:10.3389/feart.2019.00318
- 2019 McGrath, D., Webb, R., Shean, D., Bonnell, R., **H.P. Marshall**, Painter, T., Molotch, N., Elder, K., Hiemstra, C., and Brucker, L. (2019). Spatially extensive ground-penetrating radar snow depth observations during NASA’s 2017 SnowEx Campaign: Comparison to in situ, airborne, and satellite observations. *Water Resources Research*, 55(11):10026–10036, doi:10.1029/2019WR024907
- Lievens, H., Demuzere, M., **H.P. Marshall**, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Giroto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Sch ber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. *Nature Communications*, 10(1):1–12
- Lewis, G., Osterberg, E., Hawley, R., Marshall, H. P., Meehan, T., Graeter, K., McCarthy, F., Overly, T., Thundercloud, Z., and Ferris, D. (2019a). Recent precipitation decrease across the western greenland ice sheet percolation zone. *The Cryosphere*, 13:2797–2815, doi:10.5194/tc-13-2797-2019
- 2018 Graeter, K. A., Osterberg, E. C., Ferris, D. G., Hawley, R. L., **H.P. Marshall**, Lewis, G., Meehan, T., McCarthy, F., and Birkel, S. D. (2018). Ice core records of West Greenland surface melt and climate forcing. *Geophysical Research Letters*, 45(7):3164–3172, doi:10.1002/2017GL076641

Hedrick, A., Marks, D., Kormos, P., **H.P. Marshall**, Havens, S., Bormann, K., and Painter, T. H. (2018). Direct insertion of NASA Airborne Snow Observatory-derived snow depth time-series into the iSnobal energy balance snow model. *Water Resources Research*, 54(10):8045–8063, doi.org/10.1029/2018WR023190

McGrath, D., Sass, L., O’Neel, S., McNeil, C., Candela, S. G., Baker, E. H., and **H.P. Marshall** (2018). Interannual snow accumulation variability on glaciers derived from repeat, spatially extensive ground-penetrating radar surveys. *The Cryosphere*, 12:3617–3633, doi.org/10.5194/tc-12-3617-2018

Hagenmuller, P., van Herwijnen, A., Pielmeier, C., and **H.P. Marshall** (2018). Evaluation of the snow penetrometer Avatech SP2. *Cold Regions Science and Technology*, 149:83–94

McNamara, J. P., Benner, S. G., Poulos, M. J., Pierce, J. L., Chandler, D. G., Kormos, P. R., **H.P. Marshall**, Flores, A. N., Seyfried, M., Glenn, N. F., and Aishlin, P. (2018). Form and function relationships revealed by long-term research in a semiarid mountain catchment. *WIREs Water*, 5(2):e1267, doi:10.1002/wat2.1267

Brucker, L., Hiemstra, C., Marshall, H.-P., Elder, K., De Roo, R., Mousavi, M., Bliven, F., Peterson, W., Deems, J., Gadowski, P., Gelvin, A., Spaete, L., Barnhart, T., Brandt, T., Burkhart, J., Crawford, C., Datta, T., Erikstrod, H., Glenn, N., Hale, K., Holben, B., Houser, P., Jennings, K., Kelly, R., Kraft, J., Langlois, A., McGrath, D., Merriman, C., Molotch, N., Nolin, A., Polashenski, C., Raleigh, M., Rittger, K., Rodriguez, C., Roy, A., Skiles, M., Small, E., Tedesco, M., Tennant, C., Thompson, A., Uhlmann, Z., Webb, R., and Wingo, M. (2018a). Nasa SnowEx’17 in situ Measurements and Ground-Based Remote Sensing. *IGARSS 2018 - 2018 IEEE International Geoscience and Remote Sensing Symposium*, pages 6266–6268, doi:10.1109/IGARSS.2018.8517777

Kim, Y., Reck, T. J., Alonso-delPino, M., Painter, T. H., **H.P. Marshall**, Bair, E. H., Dozier, J., Chattopadhyay, G., Liou, K.-N., Chang, M.-C. F., and Tang, A. (2018). A Ku-band CMOS FMCW radar transceiver for snowpack remote sensing. *IEEE Transactions On Microwave Theory and Techniques*, 66(5):2480–2494

2017 Mikesell, T., van Wijk, K., Otheim, L., **H.P. Marshall**, and Kurbatov, A. (2017). Laser ultrasound observations of mechanical property variations in ice cores. *Geosciences*, 7(3):47

Lewis, G., Osterberg, E., Hawley, R., Whitmore, B., **H.P. Marshall**, and Box, J. (2017a). Regional Greenland accumulation variability from Operation IceBridge airborne accumulation radar. *The Cryosphere*, 11(2):773

Deeb, E. J., Marshall, H.-P., Forster, R. R., Jones, C. E., Hiemstra, C. A., and Siqueira, P. R. (2017). Supporting NASA SnowEx remote sensing strategies and requirements for L-band interferometric snow depth and snow water equivalent estimation. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1395–1396, doi:10.1109/IGARSS.2017.8127224

- Kim, E., Gatebe, C., Hall, D., Newlin, J., Misakonis, A., Elder, K., Marshall, H.-P., Hiemstra, C., Brucker, L., De Marco, E., Crawford, C., Kang, D. H., and Entin, J. (2017). NASA's SnowEx campaign: Observing seasonal snow in a forested environment. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1388–1390, doi:10.1109/IGARSS.2017.8127222
- Pegau, W. S., Garron, J., Zabilansky, L., Bassett, C., Bello, J., Bradford, J., Carns, R., Courville, Z., Eicken, H., Elder, B., Eriksen, P., Lavery, A., Light, B., Maksym, T., **H.P. Marshall**, Oggier, M., Perovich, D., Pacwiardowski, P., Singh, H., Tang, D., Wiggins, C., and Wilkinson, J. (2017). Detection of oil in and under ice. *International Oil Spill Conference Proceedings*, 2017(1):1857–1876
- Brucker, L., Hiemstra, C., Marshall, H.-P., Elder, K., De Roo, R., Mousavi, M., Bliven, F., Peterson, W., Deems, J., Gadowski, P., Gelvin, A., Spaete, L., Barnhart, T., Brandt, T., Burkhart, J., Crawford, C., Datta, T., Erikstrod, H., Glenn, N., Hale, K., Holben, B., Houser, P., Jennings, K., Kelly, R., Kraft, J., Langlois, A., McGrath, D., Merriman, C., Molotch, N., Nolin, A., Polashenski, C., Raleigh, M., Rittger, K., Rodriguez, C., Roy, A., Skiles, M., Small, E., Tedesco, M., Tennant, C., Thompson, A., Tian, L., Uhlmann, Z., Webb, R., and Wingo, M. (2017). A first overview of SnowEx ground-based remote sensing activities during the winter 2016–2017. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 1391–1394, doi:10.1109/IGARSS.2017.8127223
- 2016 Evans, S., Flores, A., Heilig, A., Kohn, M., **Marshall, H.P.**, and McNamara, J. (2016). Isotopic evidence for lateral flow and diffusive transport, but not sublimation, in a sloped seasonal snowpack, Idaho, U.S.A. *Geophysical Research Letters*, 43(7):3298–3306
- Rutter, N., **Marshall, H.P.**, Tape, K., Essery, R., and King, J. (2016). Impact of spatial averaging on radar reflectivity at internal snowpack layer boundaries. *Journal of Glaciology*, 62(236):1065–1074
- Bradford, J. H., Babcock, E. L., **Marshall, H.P.**, and Dickins, D. F. (2016). Targeted reflection-waveform inversion of experimental ground-penetrating radar data for quantification of oil spills under sea ice. *Geophysics*, 81(1):WA59–WA70
- 2015 Heilig, A., Mitterer, C., Schmid, L., Wever, N., Schweizer, J., **Marshall, H.P.**, and Eisen, O. (2015). Seasonal and diurnal cycles of liquid water in snow-measurements and modeling. *Journal of Geophysical Research-Earth Surface*, 120(10):2139–2154
- Kormos, P. R., McNamara, J. P., Seyfried, M. S., **Marshall, H.P.**, Marks, D., and Flores, A. N. (2015). Bedrock infiltration estimates from a catchment water storage-based modeling approach in the rain snow transition zone. *Journal of Hydrology*, 525:231–248
- Hedrick, A., **Marshall, H.P.**, Winstral, A., Elder, K., Yueh, S., and Cline, D. (2015). Independent evaluation of the SNODAS snow depth product using regional-scale LiDAR-derived measurements. *Cryosphere*, 9(1):13–23
- 2014 Kormos, P. R., Marks, D., McNamara, J. P., **Marshall, H.P.**, Winstral, A., and Flores, A. N. (2014a). Snow distribution, melt and surface water inputs to the soil in the mountain rain-snow transition zone. *Journal of Hydrology*, 519:190–204

- Walters, R. D., Watson, K. A., **Marshall, H.P.**, McNamara, J. P., and Flores, A. N. (2014). A physiographic approach to downscaling fractional snow cover data in mountainous regions. *Remote Sensing of Environment*, 152:413–425
- Havens, S., **Marshall, H.P.**, Johnson, J. B., and Nicholson, B. (2014b). Calculating the velocity of a fast-moving snow avalanche using an infrasound array. *Geophysical Research Letters*, 41(17):6191–6198
- Anderson, B. T., McNamara, J. P., **Marshall, H.P.**, and Flores, A. N. (2014). Insights into the physical processes controlling correlations between snow distribution and terrain properties. *Water Resources Research*, 50(6):4545–4563
- Tinkham, W. T., Smith, A. M. S., **Marshall, H.P.**, Link, T. E., Falkowski, M. J., and Winstal, A. H. (2014). Quantifying spatial distribution of snow depth errors from lidar using random forest. *Remote Sensing of Environment*, 141:105–115
- McCreight, J. L., Slater, A. G., **Marshall, H.P.**, and Rajagopalan, B. (2014). Inference and uncertainty of snow depth spatial distribution at the kilometre scale in the colorado rocky mountains: the effects of sample size, random sampling, predictor quality, and validation procedures. *Hydrological Processes*, 28(3):933–957
- Kormos, P. R., Marks, D., Williams, C. J., **Marshall, H.P.**, Aishlin, P., Chandler, D. G., and McNamara, J. P. (2014b). Soil, snow, weather, and sub-surface storage data from a mountain catchment in the rain-snow transition zone. *Earth System Science Data*, 6(1):165–173
- 2013 Havens, S., **Marshall, H.P.**, Pielmeier, C., and Elder, K. (2013b). Automatic grain type classification of snow micro penetrometer signals with random forests. *IEEE Transactions On Geoscience and Remote Sensing*, 51(6):3328–3335
- Eiriksson, D., Whitson, M., Luce, C. H., **Marshall, H.P.**, Bradford, J., Benner, S. G., Black, T., Hetrick, H., and McNamara, J. P. (2013). An evaluation of the hydrologic relevance of lateral flow in snow at hillslope and catchment scales. *Hydrological Processes*, 27(5):640–654
- Tinkham, W. T., Hoffman, C. M., Falkowski, M. J., Smith, A., **Marshall, H.P.**, and Link, T. E. (2013a). A methodology to characterize vertical accuracies in lidar-derived products at landscape scales. *Photogrammetric Engineering & Remote Sensing*, 79(8):709–716
- Rott, H., Nagler, T., Prinz, R., Voglmeier, K., Fromm, R., Adams, M. S., Yueh, S., Elder, K., **Marshall, H.P.**, Coccia, A., Imbembo, E., Schuttemeyer, D., and Kern, M. (2013). AlpSAR 2012-13, a field experiment on snow observations and parameter retrievals with Ku-and X-band radar. *ESA Living Planet Symposium*, ESA SP-722.2-13:92
- 2012 Mikesell, T. D., van Wijk, K., Haney, M. M., Bradford, J. H., **Marshall, H.P.**, and Harper, J. T. (2012). Monitoring glacier surface seismicity in time and space using rayleigh waves. *Journal of Geophysical Research-Earth Surface*, 117(F2):F02020
- 2011 **Marshall, H.P.**, Schweizer, J., and Birkeland, K. (2011). Some recent advances in snow and avalanche science. *Cold Regions Science and Technology*, 69(2-3):119–121

- Geroy, I., Gribb, M., **Marshall, H.P.**, Chandler, D., Benner, S. G., and McNamara, J. P. (2011). Aspect influences on soil water retention and storage. *Hydrological Processes*, 25(25):3836–3842
- 2010 Tape, K. D., Rutter, N., **Marshall, H.P.**, Essery, R., and Sturm, M. (2010b). Recording microscale variations in snowpack layering using near-infrared photography. *Journal of Glaciology*, 56(195):75–80
- Tape, K. D., Lord, R., **Marshall, H.P.**, and Ruess, R. W. (2010a). Snow-mediated ptarmigan browsing and shrub expansion in arctic alaska. *Ecoscience*, 17(2):186–193
- Koh, G., Lever, J., Arcone, S., **Marshall, H.P.**, and Ray, L. (2010). Autonomous fmcw radar survey of antarctic shear zone. *IEEE Publications, 13th International Conference on Ground Penetrating Radar*, DOI: 10.1109/ICGPR.2010.5550174
- 2009 **Marshall, H.P.** and Johnson, J. B. (2009). Accurate inversion of high-resolution snow penetrometer signals for microstructural and micromechanical properties. *Journal of Geophysical Research-Earth Surface*, 114(F04016)
- Lutz, E., Birkeland, K., and **Marshall, H.P.** (2009b). Quantifying changes in weak layer microstructure associated with artificial load changes. *Cold Regions Science and Technology*, 59(2-3):202–209
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- Youngblood, P. and **H.-P. Marshall** (2021). Continuous snow temperature monitoring for the idaho highway 21 avalanche program. Technical Report FHWA-ID-21-276, Idaho Transportation Department, 3311 W State Street, P. O. Box 7129, Boise, ID 83707-1129 United States
- 2019 Meehan, T. G., **H.-P. Marshall**, Deeb, E. J., and Shoop, S. A. (2019). Snowmicropenetrometer applications for winter vehicle mobility. ERDC 6.2 Boreal Aspects of Ensured Maneuver (BAEM) TR-19-14, U.S. Army Cold Regions Research and Engineering Laboratory
- 2015 **Marshall, H.P.**, Hawley, R., and Tedesco, M. (2015). *Field measurements for remote sensing of the cryosphere*, chapter 14, pages 345–381. Remote Sensing of the Cryosphere. John Wiley & Sons, Ltd, ed. M. Tedesco
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- 2014 Havens, S., **Marshall, H.P.**, Trisca, G., and Johnson, J. B. (2014c). Real time avalanche detection for high risk areas. Monograph FHWA-ID-14-219, Idaho Transportation Department, 3311 W State Street, P. O. Box 7129, Boise, ID 83707-1129 United States

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- 2021 Brucker, L., Vuyovich, C. M., Elder, K., **H.-P. Marshall**, and Hiemstra, C. (2021). Snowex20 grand mesa autumn 2019 snow pits, version 1. Technical Report doi: <https://doi.org/10.5067/4IZGCQC1J31Q>, NASA National Snow and Ice Data Center Distributed Active Archive Center

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- Mason, M., **H.-P. Marshall**, and Merkouriadi, I. (2020). SnowEx20 SnowMicroPen (SMP) Raw Penetration Force Profiles, Version 1. Technical Report doi: <https://doi.org/10.5067/ZYW6IHFRYDSE>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- Johnson, J., Anderson, J., **H.-P. Marshall**, Havens, S., and Watson, L. (2020). Dataset on fortress snow avalanche 13 march 2018. *Boise State University Infrasound Repository*, 4
- 2018 Elder, K., Brucker, L., Hiemstra, C., and **H.P. Marshall** (2018). Snowex17 community snow pit measurements, version 1. Digital media, <https://doi.org/10.5067/Q0310G1XULZS>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- 2018 Brucker, L., Hiemstra, C., **H.P. Marshall**, and Elder, K. (2018b). Snowex17 community snow depth probe measurements, version 1. Digital media, <https://doi.org/10.5067/WKC6VFMT7JTF>, NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado USA
- 2004 **Marshall, H.P.**, Koh, G., and Forster, R. (2004a). CLPX-Ground: Ground-based Frequency Modulated Continuous Wave (FMCW) radar. Digital media, <http://nsidc.org/data/nsidc-0164.html>, National Snow and Ice Data Center, University of Colorado at Boulder

Magazine Articles / Podcasts

- 2013 Durham, T., **Marshall, H.P.**, Tsang, L., Racette, P., Bonds, Q., Miranda, F., and Vanhille, K. (2013). Wideband sensor technologies for measuring surface snow. *Earthzine*, December 2(earthzine.org)

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| 2012 | Marshall, H.P. (2012). Researching snow: More than just hitting the slopes. Podcast https://beyondtheblue.boisestate.edu/blog/2012/01/09/hp-marshall/ , Beyond the Blue: Boise State Faculty Podcasts |
| 2011 | Sturm, M. and Marshall, H.P. (2011). Marshall receives Cryosphere Young Investigator Award (citation and response). <i>EOS Transactions, AGU</i> , 92(22):189 |

Volunteer Activities

Special Olympics basketball coach (02/15-06/15)
ESL Teacher for refugees (08/09-07/10)
Mentor, Big Brothers / Big Sisters of La Plata County (07/2006-08/2008)
Volunteer high school teacher, Old Crow, Yukon Territory, Canada (1999-2000)
Helped install seismic station on the summit of Mt. Rainier (1999)
Logan School of Learning (2-3rd graders), Field Trip leader to Loveland Pass, CO (2003)
Co-taught 3-day English course to indigenous leaders in southern Chile (2003)
Guest lecturer, Nathan Hale High School Physics class (1999)
Notetaker for Disabled Student Services, University of Washington (1996)
Teen Feed (1995)

Affiliations

Member, Snow Radar Algorithm Team, NASA SCLP (2004-present)
International Glaciological Society (IGS)
American Geophysical Union (AGU, Cryosphere Representative 2008-2012)
Sigma Xi (Full member)
American Avalanche Association (Research Chair 2005-2011, Professional Member)
American Institute for Avalanche Research and Education (AIARE) Certified Instructor (2003-04)
American Meteorological Society (Full Member)

Interests / Hobbies

Backcountry skiing / snowboarding, mountaineering, rock climbing, basketball, trail running, mountain biking
West African percussion, Flamenco percussion, yoga